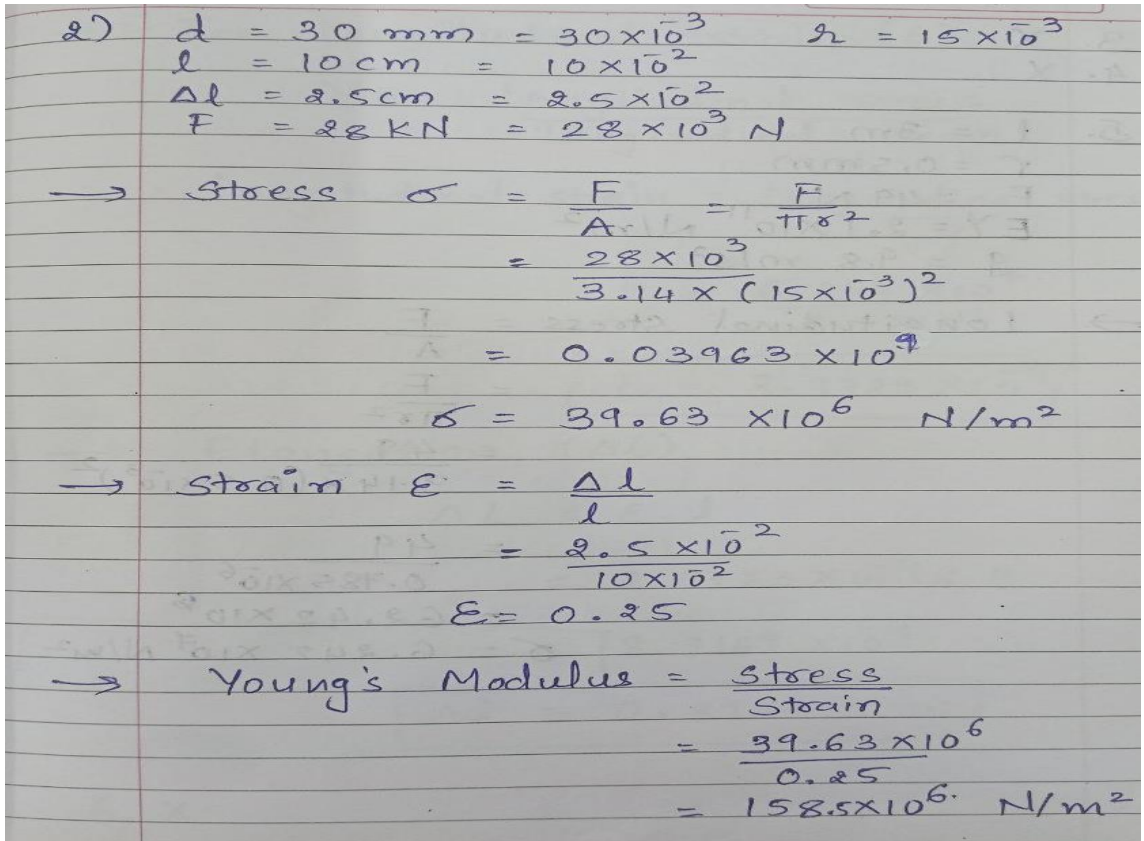
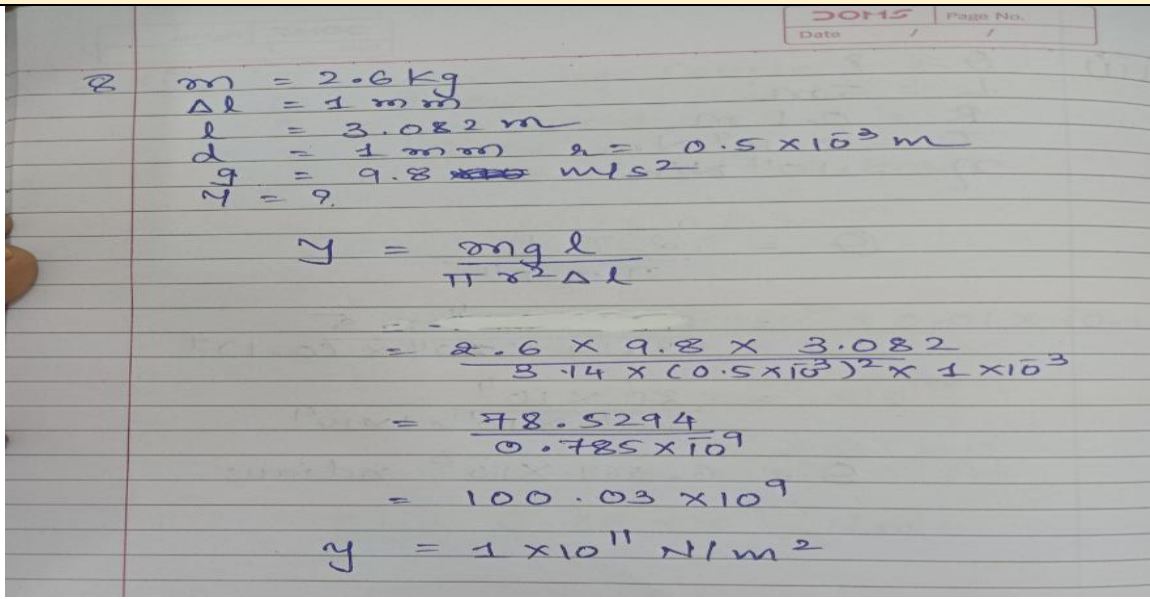
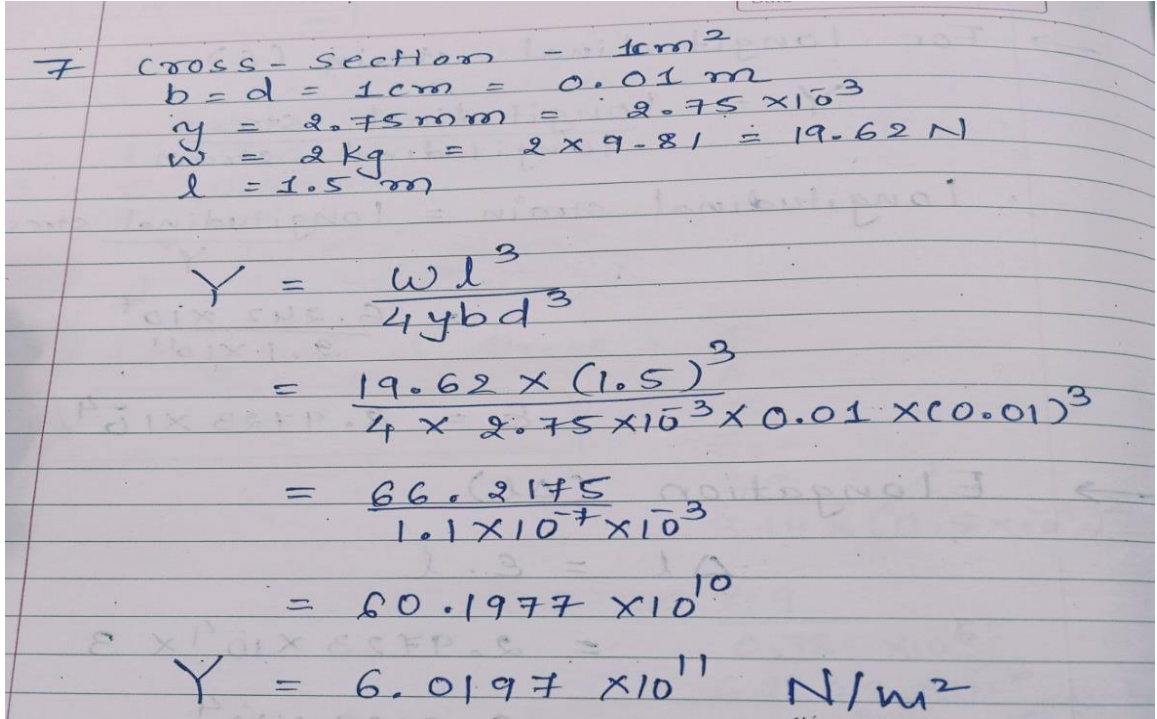
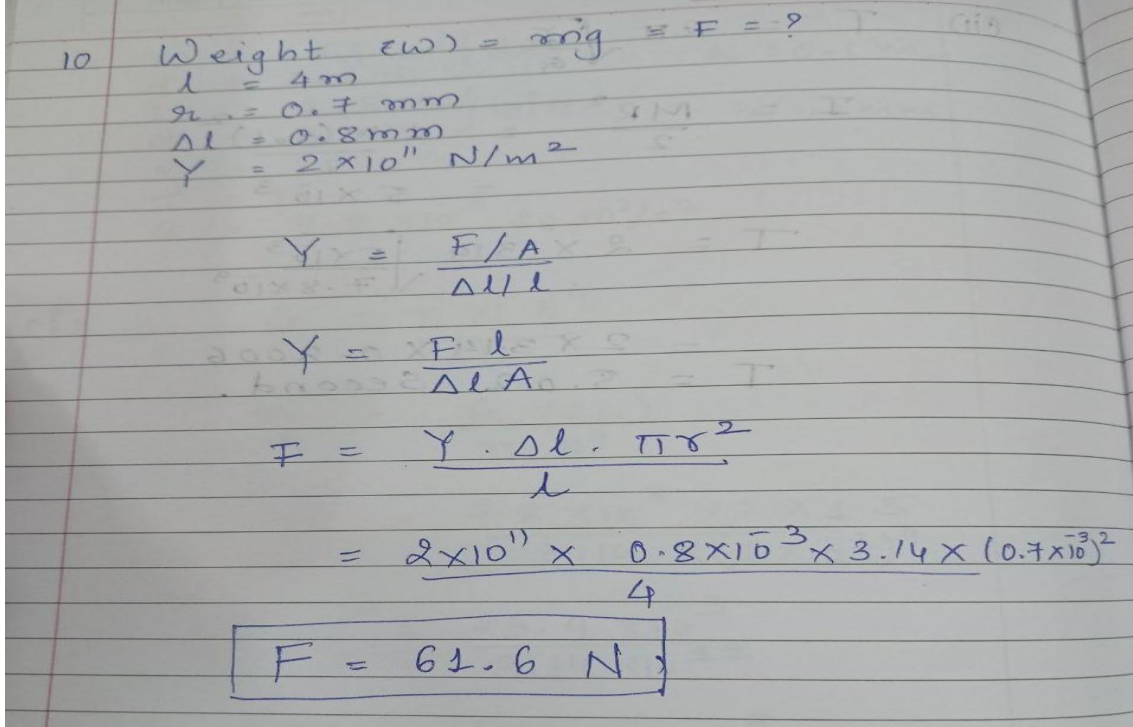


UNIT-1

EXAMPLES

1.	An elastic rod having diameter of 30 mm, 10 cm long extends by 2.5 cm under tensile load of 28 kN. Find the stress, strain and the Young's modulus for the material of the rod. (Jun-2019) (Jan-2024) [NLJIET]	3
Sol	 2) $d = 30 \text{ mm} = 30 \times 10^{-3}$ $l = 10 \text{ cm} = 10 \times 10^{-2}$ $\Delta l = 2.5 \text{ cm} = 2.5 \times 10^{-2}$ $F = 28 \text{ kN} = 28 \times 10^3 \text{ N}$ $\rightarrow \text{Stress } \sigma = \frac{F}{A} = \frac{F}{\pi r^2}$ $= \frac{28 \times 10^3}{3.14 \times (15 \times 10^{-3})^2}$ $= 0.03963 \times 10^9$ $\sigma = 39.63 \times 10^6 \text{ N/m}^2$ $\rightarrow \text{Strain } \epsilon = \frac{\Delta l}{l}$ $= \frac{2.5 \times 10^{-2}}{10 \times 10^{-2}}$ $\epsilon = 0.25$ $\rightarrow \text{Young's Modulus} = \frac{\text{Stress}}{\text{Strain}}$ $= \frac{39.63 \times 10^6}{0.25}$ $= 158.5 \times 10^6 \text{ N/m}^2$	
2.	A load of 2.6 kg produces an extension of 1 mm in a wire of 3.082 m and 1 mm in diameter. Calculate the Young's modulus of a wire. (Mar-2021) [NLJIET]	3
Sol	 2) $m = 2.6 \text{ kg}$ $\Delta l = 1 \text{ mm}$ $l = 3.082 \text{ m}$ $d = 1 \text{ mm}$ $r = 0.5 \times 10^{-3} \text{ m}$ $g = 9.8 \text{ m/s}^2$ $Y = ?$ $Y = \frac{mg l}{\pi r^2 \Delta l}$ $= \frac{2.6 \times 9.8 \times 3.082}{3.14 \times (0.5 \times 10^{-3})^2 \times 1 \times 10^{-3}}$ $= \frac{78.5294}{0.785 \times 10^{-9}}$ $= 100.03 \times 10^9$ $Y = 1 \times 10^{11} \text{ N/m}^2$	

3.	A steel wire has 3 m length and 0.5 mm radius. When it is stretched by force of 49 N find (i) longitudinal stress, (ii) longitudinal strain and (iii) elongation produced in the wire if Young's modulus of steel is $2.1 \times 10^{11} \text{ N/m}^2$. Take $g = 9.8 \text{ m/s}^2$. (Aug-2022)[NLJIET]	3
Sol	5. $l = 3\text{m}$ $r = 0.5\text{mm}$ $F = 49\text{N}$ $Y = 2.1 \times 10^{11} \text{ N/m}^2$ $g = 9.8 \text{ m/s}^2$ → Longitudinal stress = $\frac{F}{A}$ $= \frac{F}{\pi r^2}$ $= \frac{49}{3.14 \times (0.5 \times 10^{-3})^2}$ $= \frac{49}{0.785 \times 10^{-6}}$ $= 62.42 \times 10^8$ $\sigma = 6.242 \times 10^7 \text{ N/m}^2$ → For longitudinal strain (ϵ) $Y = \frac{\text{longitudinal stress}}{\text{longitudinal strain}}$ ∴ Longitudinal strain = $\frac{\text{longitudinal stress}}{Y}$ $= \frac{6.242 \times 10^7}{2.1 \times 10^{11}}$ $\epsilon = 2.9723 \times 10^{-4}$ → Elongation (Δl) $\Delta l = \epsilon \cdot l$ $= 2.9723 \times 10^{-4} \times 3$ $= 8.9169 \times 10^{-4}$ $\Delta l = 0.8917 \text{ mm}$	

4.	A brass bar having a cross-section of 1 cm^2 is supported on two knife-edges 1.5 m apart. A load of 2 kg at the center of the bar depresses that point by 2.75 mm. What is the Young's Modulus for brass? (Mar-2022) [NLJIET]	3
	 Handwritten solution for Question 4: 7 cross-section = 1 cm^2 $b = d = 1 \text{ cm} = 0.01 \text{ m}$ $y = 2.75 \text{ mm} = 2.75 \times 10^{-3}$ $w = 2 \text{ kg} = 2 \times 9.81 = 19.62 \text{ N}$ $l = 1.5 \text{ m}$ $Y = \frac{wl^3}{4ybd^3}$ $= \frac{19.62 \times (1.5)^3}{4 \times 2.75 \times 10^{-3} \times 0.01 \times (0.01)^3}$ $= \frac{66.2175}{1.1 \times 10^{-7} \times 10^{-3}}$ $= 60.1977 \times 10^{10}$ $Y = 6.0197 \times 10^{11} \text{ N/m}^2$	
5.	An unknown weight is attached to the lower end of wire of length 4 m, radius 0.7 mm, extends it by 0.8 mm. If $Y = 2 \times 10^{11} \text{ N/m}^2$. Find the unknown weight. (Jan-2025-New) [NLJIET]	3
	 Handwritten solution for Question 5: 10 Weight (w) = $mg = F = ?$ $l = 4 \text{ m}$ $r = 0.7 \text{ mm}$ $\Delta l = 0.8 \text{ mm}$ $Y = 2 \times 10^{11} \text{ N/m}^2$ $Y = \frac{F/A}{\Delta l/l}$ $Y = \frac{F \cdot l}{\Delta l \cdot A}$ $F = \frac{Y \cdot \Delta l \cdot \pi r^2}{l}$ $= \frac{2 \times 10^{11} \times 0.8 \times 10^{-3} \times 3.14 \times (0.7 \times 10^{-3})^2}{4}$ $F = 61.6 \text{ N}$	

6.

A solid disc of 1 kg mass and 0.2 m diameter is suspended in a horizontal plane by a vertical wire attached to its center. The length and diameter of the wire is 1.5 m and 2 mm, respectively. Calculate modulus of rigidity of wire and the time period Torsional oscillations if Torsional rigidity $CS = 7.8 \times 10^{-3} \text{ kg} \cdot \text{m}^2 / \text{s}^2$. (Jun-2019) [NLJIET]

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12 $m = 1 \text{ kg}$ } disc
 $d = 0.2 \text{ m}$ }
 $L = 1.5 \text{ m}$ } wire
 $d = 2 \text{ mm}$ }
 $\eta = ?$
 $T = ?$
 $CS = 7.8 \times 10^{-3} \text{ kg} \cdot \text{m}^2 / \text{s}^2$

(i) $C = \frac{\pi \eta R^4}{2L}$

$\eta = \frac{C \cdot (2L)}{\pi R^4}$

$= \frac{7.8 \times 10^{-3} \times 2 \times 1.5}{3.14 \times (1 \times 10^{-3})^4}$

$= \frac{23.4 \times 10^{-3}}{3.14 \times 10^{-12}}$

$\eta = 7.45 \times 10^9 \text{ N/m}^2$

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(ii) $T = 2\pi \sqrt{\frac{I}{C}}$

$I = \frac{MR^2}{2} = \frac{1 \times (0.1)^2}{2}$

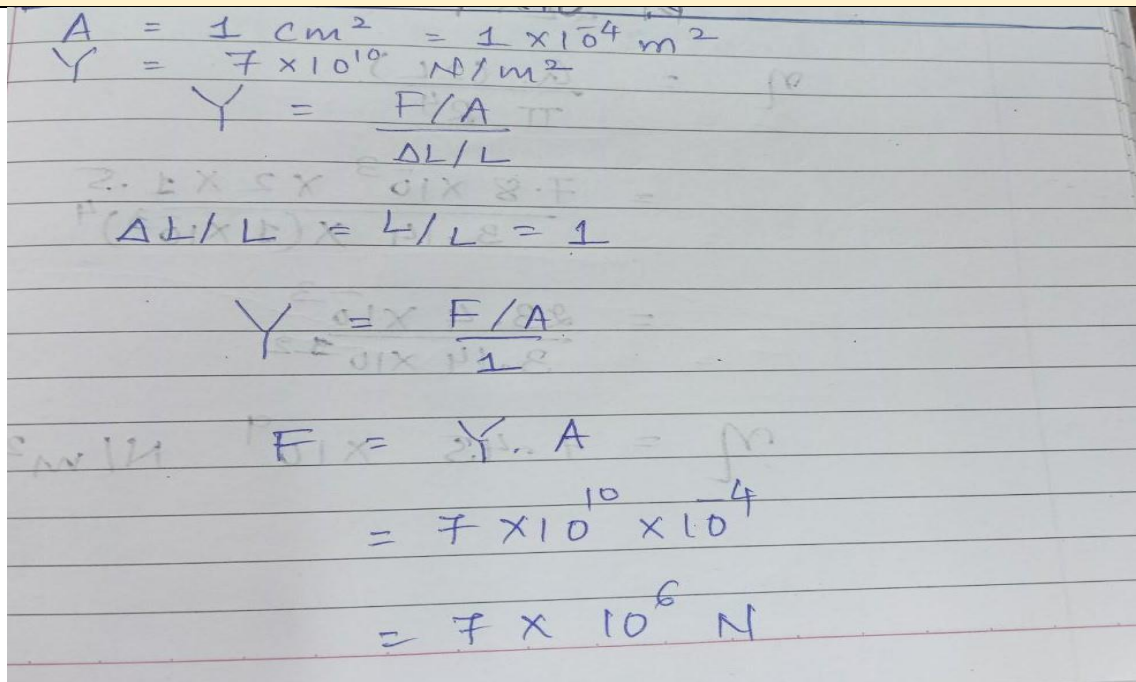
$= 5 \times 10^{-3}$

$T = 2 \times 3.14 \sqrt{\frac{5 \times 10^{-3}}{7.8 \times 10^9}}$

$= 2 \times 3.14 \times 0.0008006$

$T = 5.02 \text{ Second.}$

7.	(ii) What force is required to stretch a steel wire to double the length when its area of cross section is 1 cm^2 . Given that Young's modulus of wire is $7 \times 10^{10} \text{ N/m}^2$. (Jan-2020) [NLJIET]	4
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Handwritten solution for Question 7:

$$A = 1 \text{ cm}^2 = 1 \times 10^{-4} \text{ m}^2$$

$$Y = 7 \times 10^{10} \text{ N/m}^2$$

$$Y = \frac{F/A}{\Delta L/L}$$

$$(\Delta L/L) = L/L = 1$$

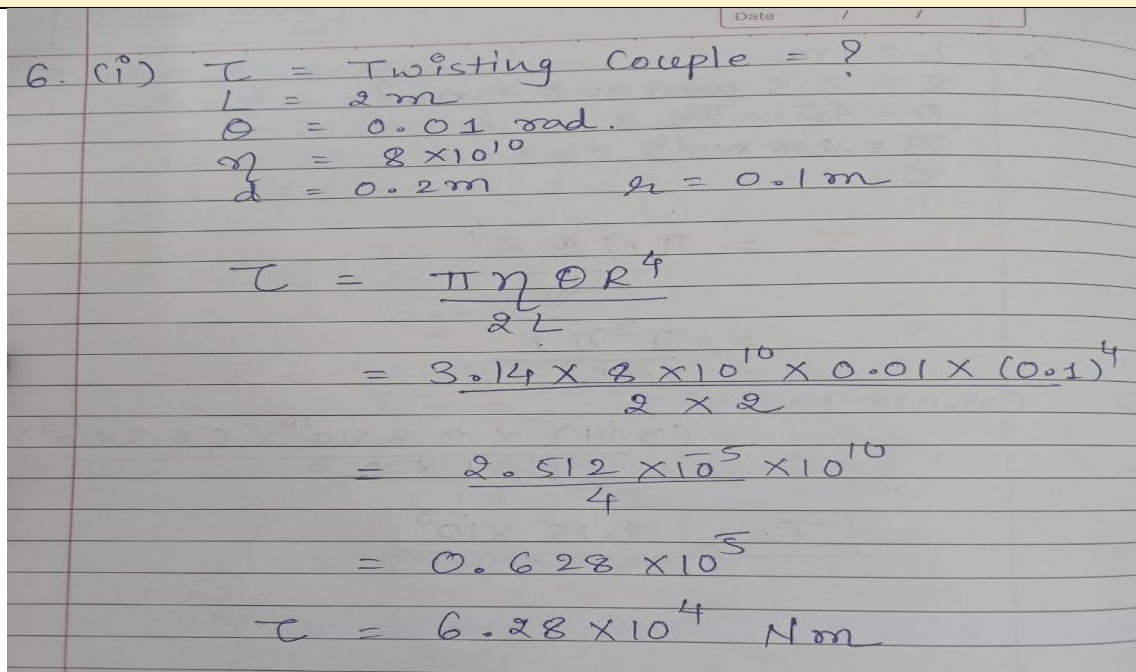
$$Y = \frac{F/A}{1}$$

$$F = Y \cdot A$$

$$= 7 \times 10^{10} \times 10^{-4}$$

$$= 7 \times 10^6 \text{ N}$$

8.	(i) Calculate the twisting couple of a solid shaft of 2 m length and 0.2 m diameter when it is twisted through an angle of 0.01 rad. The coefficient of rigidity of the material is $8 \times 10^{10} \text{ N/m}^2$. (ii) Calculate the twist produced in a solid wire of 5 m length and 0.1 m radius. The twisting couple produced in the wire is $2 \times 10^4 \text{ N m}$ and the coefficient of rigidity of the material is 10^{11} N/m^2 . (Aug-2022) [NLJIET]	4
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Handwritten solution for Question 8:

6. (i) $T = \text{Twisting Couple} = ?$
 $L = 2 \text{ m}$
 $\theta = 0.01 \text{ rad.}$
 $\eta = 8 \times 10^{10}$
 $d = 0.2 \text{ m} \quad r = 0.1 \text{ m}$

$$T = \frac{\pi \eta \theta R^4}{2L}$$

$$= \frac{3.14 \times 8 \times 10^{10} \times 0.01 \times (0.1)^4}{2 \times 2}$$

$$= \frac{2.512 \times 10^5 \times 10^{10}}{4}$$

$$= 0.628 \times 10^5$$

$$T = 6.28 \times 10^4 \text{ Nm}$$

(ii)

$$\theta = ?$$

$$L = 5 \text{ m}$$

$$R = 0.1 \text{ m}$$

$$T = 2 \times 10^4 \text{ Nm}$$

$$\eta = 10^{11} \text{ N/m}^2$$

$$\theta = \frac{2 T L}{\pi \eta R^4}$$

$$= \frac{2 \times 2 \times 10^4 \times 5}{3.14 \times 10^{11} \times (0.1)^4}$$

$$= \frac{20 \times 10^4}{3.14 \times 10^{11} \times 1 \times 10^{-4}}$$

$$\theta = 6.369 \times 10^{-3} \text{ radians.}$$

9. A wire of length 2 m and radius 0.5 mm is clamped at one of its ends. Calculate the torque required to twist the other end by 90° . The rigidity modulus of the wire is $2.8 \times 10^{10} \text{ N/m}^2$. (Mar-2023) [NLJIET]

4

N

$$L = 2 \text{ m}$$

$$R = 0.5 \text{ mm} = 0.5 \times 10^{-3}$$

$$\theta = 90^\circ = \pi/2$$

$$\eta = 2.8 \times 10^{10}$$

$$T = ?$$

$$T = \frac{\pi \eta \theta R^4}{2L}$$

$$= \frac{\pi^2 \eta R^4}{4L}$$

$$= \frac{(3.14)^2 \times 2.8 \times 10^{10} \times (0.5 \times 10^{-3})^4}{4 \times 2}$$

$$T = 2.15 \times 10^{-3} \text{ Nm}$$

10. Find the force required to increase the length of a wire of 10^{-6} m^2 area of cross-section by 50 % whose Young's modulus is $2 \times 10^{11} \text{ N/m}^2$.
(Jun-2025-New) [NLJIET]

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10 Area of cross-section $A = 10^{-6} \text{ m}^2$

Δl = Extension in length = 50 % of l

$$\frac{\Delta l}{l} = \frac{50}{100} = 0.5 = \text{strain}$$

$$Y = 2 \times 10^{11} \text{ N/m}^2 \quad F = ?$$

$$\rightarrow Y = \frac{\text{Stress}}{\text{strain}}$$

$$\rightarrow \therefore \text{Stress} = Y \cdot \text{Strain}$$

$$= 2 \times 10^{11} \times 0.5$$

$$= 1 \times 10^{11} \text{ N/m}^2$$

$$\rightarrow \text{Force} = \text{Stress} \times A$$

$$= 1 \times 10^{11} \times 10^{-6}$$

$$\boxed{F = 1 \times 10^5 \text{ N}}$$